HEATMAP NODE · FEASIBILITY

HeatMap Node feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First feasibility assessment at the kit 1.9.1 upgrade, from HMN-CAL-001, HMN-STM-001 and the D3 model

HeatMap Node feasibility assessment

Verdict: option A is feasible on paper at TRL 3 as a garage-built sensor head (USD 136 against a USD 130 target) on the FieldNode core, with two open measurement questions (the street error of the derived WBGT and the fan flow) that only a collocated field comparison can close at TRL 4; option B is feasible in principle but depends on a water upkeep cycle that few partners will keep. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Of 40 STEM rows, 21 are within known science and 15 plausible; three failures were fixed and one (the FieldNode cell charge limit) goes to Amish; the D3 models of both options pass `cad_check` with no interference	High	WBGT model error for streets unknown (PR-005 at risk); sensors at 2.7 m (PR-009 not met)	HMN-STM-001, HMN-CAL-001
Materials	Aluminium 6063 and 5052, 316 stainless, copper globe, ASA prints, EPDM trim; no exotic material	Medium	ASA heat deflection of the chosen filament, stainless on aluminium corrosion and paint ageing are plausible, not shown (S22 to S24)	HMN-STM-001 S22 to S24
Manufacturing and assembly	Saddle formed from 3 mm sheet, shield and cowl printed, arm drilled; fastened with 63 fasteners in the D3 model; build plan with pictures	Medium	Forming the saddle in a garage needs a brake or a sheet-metal shop; strap-band preload (S11)	HMN-BLD-001 (`docs/05-build-plan.md`), `cad/fidelity.json`
Power and controls	Sensors and fan 35.4 mW, 35 % of FieldNode's 100 mW allowance; 5.75 days without sun	Medium	Cell charging above its rated temperature (S25, PR-016) is a FieldNode item	FND-CAL-001, HMN-CAL-001 F2
Supply of parts	All bought parts are catalogue items (SHT45, NTC, fan, cup anemometer, M12 leads)	Medium	Prices indicative; a cup anemometer of the assumed start-up is not chosen	`bom/bom.csv` , HMN-MKT-001
Cost	USD 136.00 for the head, USD 6 over the USD 130 target; full node USD 292.00	Medium	Savings listed in HMN-DEC-001 (one 8-pin lead about USD 4)	`bom/bom.csv` , PR-013
Regulatory	No personal data; radio inside FieldNode's grant; pole permission per city	Medium	Module certification and pole standards to assess	HMN-REG-001
Field operation	Visits about every 6 months for option A; 5 to 7 days for option B	Medium (A), Low (B)	Insects in shield gaps and fan (S27); wick upkeep (S35)	HMN-STM-001

Ratings: High, Medium, Low.